

OOS Impact Analytics — Stock-out Detection Methodology

Lambda / Poisson logic and a worked example (VV-VITA-208). Prepared for review.

1. The question

For each SKU and marketplace we observe daily units sold. A day with **zero sales** may be a genuine stock-out (lost sales) or just a normal quiet day. We need a rule that tells these apart without manual review, then values the lost units in EUR (revenue and contribution margin).

2. The lambda / Poisson rule

Let **lambda** be a SKU's expected demand rate = its trailing 90-day average units per calendar day (per marketplace, forward-filled). Sales arriving through a day behave approximately like a Poisson process.

Background — the Poisson distribution. For a process expecting lambda events in a given interval, the probability of exactly k events in that interval is:

$$P(k) = (\lambda^k \times e^{-\lambda}) / k!$$

Classic uses include the number of calls received at a call centre, or radioactive-decay events in a fixed observation period. For instance, a call centre averaging lambda = 3 calls per minute receives 1 to 4 calls in any given minute with probability about 0.77, while 0 or at least 5 calls has probability about 0.23. Our SKUs are directly analogous: units sold per day in place of calls per minute.

Setting k = 0 (a zero-sales day), the formula reduces to what we use:

$$P(0 \text{ sales}) = e^{-\lambda}$$

So the higher a SKU's demand rate, the less plausible a zero-sales day is by chance — and the stronger the signal that something blocked availability.

We flag a zero-sales day as a (marketplace) stock-out when lambda clears a threshold (default **3 units/day**), in addition to two guards: the SKU must be an established seller (sold before) and the zero-run must be enclosed by later sales or be ongoing in the last 21 days (so we ignore pre-launch and discontinued products). Physical stock-outs (EU sellable balance = 0 in the Amazon ledger) are always flagged regardless.

How the threshold maps to false-positive rate:

lambda (units/day)	P(0)=e ^{-lambda}	Reading
1	~37%	zeros are normal -> ignore
2	~14%	still common
3 (default)	~5%	zero is anomalous -> flag
5	~0.7%	near-certain block
30	~0%	unmistakable

Why 3 as the default: it is a reasoned balance, not a proven optimum. P(0)=e⁻³ is about 5%, the conventional ~2-sigma significance cutoff, so a zero day below ~5% natural likelihood counts as a real anomaly. Below 3 the false-positive rate climbs steeply (14% at 2, 37% at 1) and the slow-moving long tail floods the results; much above 3 we start missing genuine stock-outs of moderate-velocity SKUs. On the live data 3/day cleanly excluded thin-market noise (FR/IT at ~1/day) while catching the real DE/UK events, and produced a sensible total (about EUR 238k lost CM3/yr; median ~10 OOS days per affected SKU). It is exposed as a slider so reviewers can tune sensitivity; a per-SKU k-sigma rule (flag when lambda >= k x std) is the natural next step.

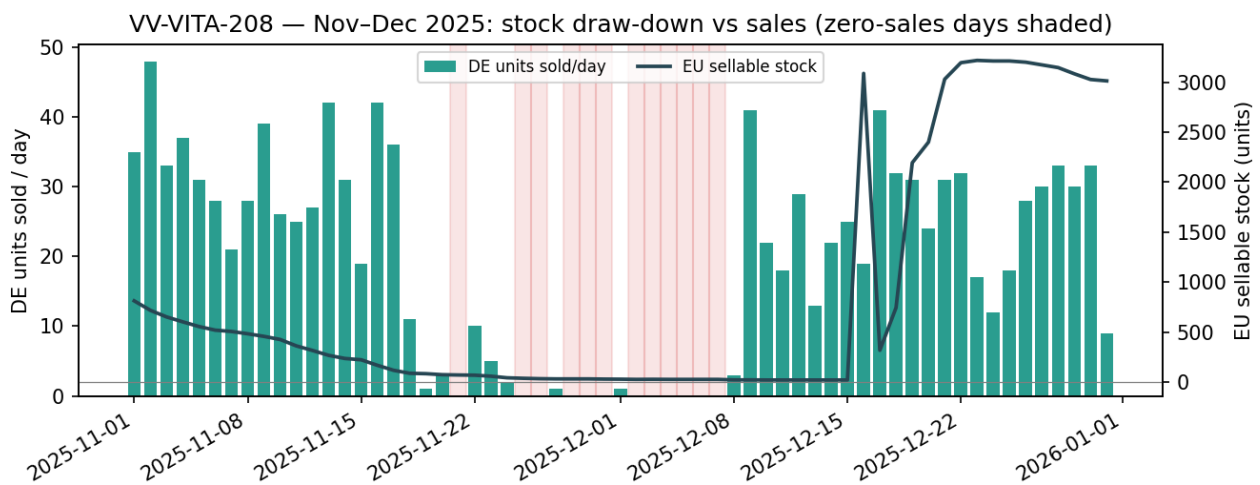
3. Worked example — VV-VITA-208 across marketplaces

Same SKU, very different verdicts, driven entirely by lambda (approx. yearly average units/day):

Marketplace	Units / yr	lambda (units/day)	$P(0)=e^{(-\lambda)}$	Flag @ 3/day?
amazon.de	12,517	34.3	0.0%	Yes
amazon.co.uk	1,942	5.3	0.5%	Yes
amazon.es	719	2.0	13.9%	No
amazon.fr	414	1.1	32.2%	No
amazon.it	381	1.0	35.2%	No

In Germany a zero-sales day is essentially impossible by chance, so it is a true stock-out; in France/Italy (~1/day) zeros happen ~1 day in 3 and carry no signal. This is why the model flagged this SKU in DE and UK but not FR/IT/ES.

4. The real Nov-Dec 2025 stock-out (from the Amazon ledger)



EU network stock fell from 814 to about 30 units across November for a SKU selling 30-50/day in DE; no replenishment arrived until 16 Dec (a 3,087-unit receipt). As stock ran critically low, DE sales flatlined for 12 days (red). Crucially the EU balance never hit zero (floor ~23), so a pure 'balance = 0' rule would have caught none of these days — the lambda/Poisson marketplace signal is what detects this low-stock throttle. Estimated lost units on each flagged day = lambda (the current demand rate), valued at the SKU's trailing average price and CM3 per unit.

Caveat: Novadata sales and Amazon ledger shipments do not align perfectly day-to-day (order vs fulfilment dating), and real sales are somewhat more variable than pure Poisson (weekends, promotions). The percentages above are decision intuition, not exact guarantees; the threshold is tunable.